

Computer Security

Tutorial Questions (60 Marks)

Topic: Security Policies & Security Standards

Time Allowed: 1 Hour

Based on Lecture 7 and Lecture 8 slides.

Section A: Scenario-Based Short Questions- (30 Marks)

Question 1 (5 Marks)

A university discovered that many students were sharing passwords with friends to access online learning systems. The university already had a security policy stating:

“Access to network resources will be granted through a unique user ID and password.”

However, no technical enforcement existed.

Required:

- a) Identify ONE weakness in the university’s current approach. (1 mark)
- b) Suggest TWO suitable security standards that can support the policy. (2 marks)
- c) Explain why policies alone are insufficient without standards. (2 marks)

Question 2 (5 Marks)

A company introduced the following password rule:

- Passwords must contain upper-case letters, numbers, and special characters.
- Passwords expire every 90 days.

Many employees complained that the rules were difficult to follow, and management became less strict in enforcement.

Required:

- a) Classify whether the above statements are policies, standards, or guidelines. (2 marks)
- b) Explain ONE risk if management support becomes weak. (1 mark)
- c) Suggest ONE awareness activity that can improve compliance among employees. (2 marks)

Question 3 (5 Marks)

A hospital recently installed a new cloud-based patient management system. During deployment, the IT team realized there was no existing security policy regarding cloud storage and remote access.

Required:

- a) Explain why new technologies may require policy modification. (2 marks)
- b) Identify TWO risks the hospital should assess before deployment. (2 marks)
- c) State ONE possible consequence if policies are not updated. (1 mark)

Question 4 (5 Marks)

A bank plans to apply for ISO/IEC 17799 accreditation to improve customer trust. However, senior management argues that the process is expensive and unnecessary.

Required:

- a) State ONE advantage of ISO/IEC 17799 accreditation. (1 mark)
- b) Explain why management support is important in security implementation. (2 marks)
- c) Suggest TWO reasons why organisations fail to adopt security standards. (2 marks)

Question 5 (5 Marks)

An organisation installed advanced firewalls and antivirus systems but never reviewed logs, trained staff, or monitored policy compliance.

Six months later, attackers successfully stole sensitive information using phishing emails.

Required:

- a) Identify ONE human-related weakness in the organisation's security management. (1 mark)
- b) Explain why technical controls alone cannot provide 100% security. (2 marks)
- c) Suggest TWO additional security management activities that should have been implemented. (2 marks)

Question 6 (5 Marks)

A retail company allowed employees to use personal USB drives on office computers without restrictions. Malware later spread across the company network.

Required:

- a) Suggest ONE security policy that could reduce this risk. (2 marks)
- b) Suggest ONE technical standard to enforce the policy. (1 mark)
- c) Explain the relationship between policy and standard using this scenario. (2 marks)

Section B: Logic & Analytical Questions (30 Marks)

Question 7 (10 Marks)

A university plans to strengthen its cybersecurity governance framework.

The management team proposes the following actions:

1. Create password complexity requirements
2. Train staff on phishing awareness
3. Define acceptable use of university computers
4. Install biometric access systems
5. Conduct regular internal audits

Required:

Classify each action into the MOST appropriate category below and justify your answer:

- Policy
- Standard
- Guideline
- Awareness
- Control/Monitoring

(2 marks each)

Question 8 (10 Marks)

Read the scenario below carefully.

A multinational company has offices in different countries. Each branch independently creates its own security rules. As a result:

- Password requirements differ between branches.
- Some branches encrypt data while others do not.
- Employees transferring between branches become confused.
- Security incidents increase.

Required:

- a) Explain THREE problems caused by inconsistent security standards. (6 marks)
- b) Explain how baseline security can help the organisation. (2 marks)
- c) Suggest TWO security areas that should be standardized across all branches. (2 marks)

Question 9 (10 Marks)

An organisation conducted a risk assessment and identified the following:

- Critical customer database
- Weak password practices
- Lack of employee awareness
- No disaster recovery plan

Management only approved purchasing new antivirus software.

Required:

- a) Critically evaluate whether management made the correct decision. (4 marks)
- b) Identify TWO additional controls the organisation should prioritize. (2 marks)
- c) Explain why risk assessment is important before developing security policies and standards. (4 marks)

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